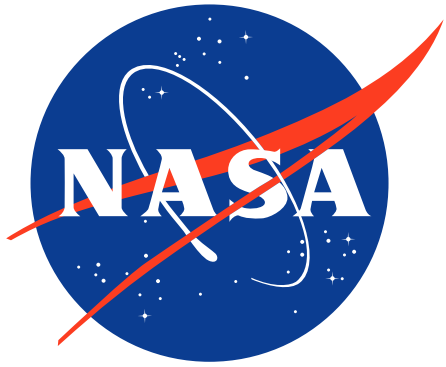




The Noise Measured by Back-illuminated Silicon Sensors

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IRIS Noise Discrepancy

- Interface Region Imaging Spectrograph (IRIS)
 - Ultraviolet (UV) solar spectrograph
 - Built and operated with the help of MSU
- IRIS CCDs measure less noise than predicted
 - UV photon transfer curve is too steep
 - Expected slope ratio > 2
 - Measured slope ratio = 1.5
 - SNR is $\sim 25\%$ better than expected

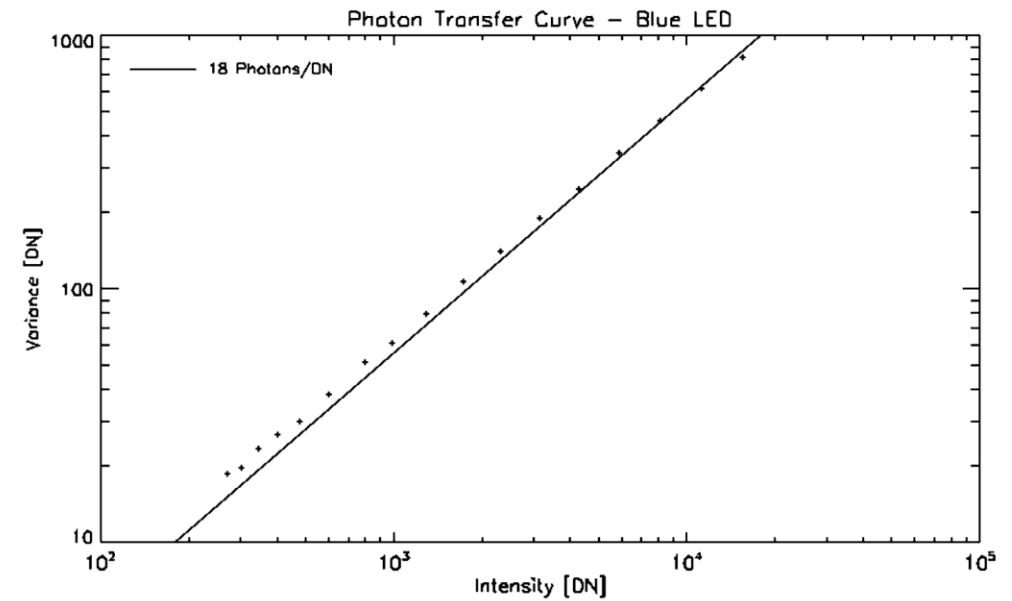


Figure 26 Photon transfer curve for the SJI CCD measure with a blue LED.

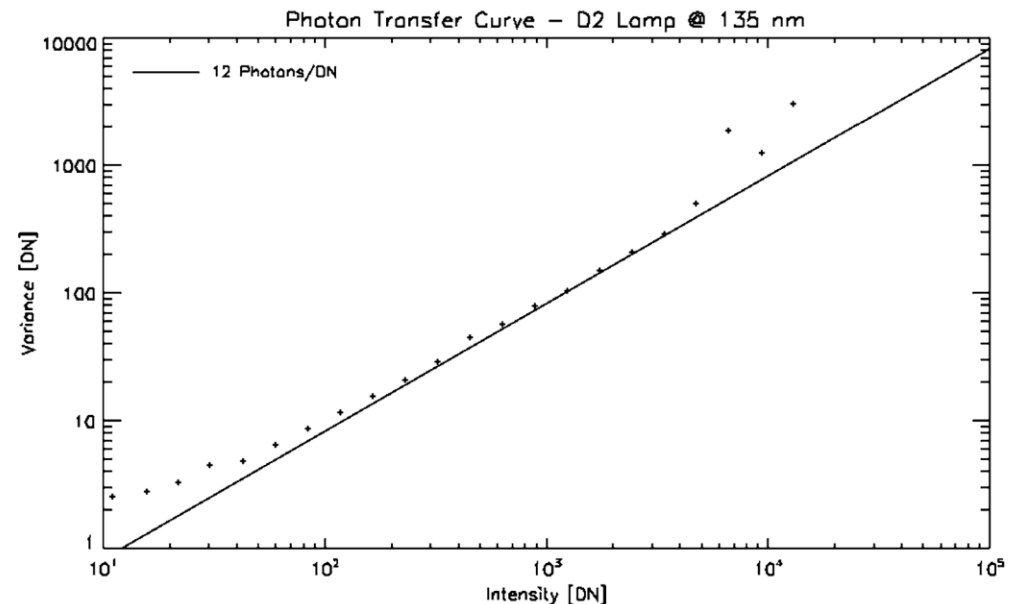


Figure 27 Photon transfer curve for the SJI CCD measured with a deuterium lamp at 1350 Å.

HST/WFC3 Noise

- Wide Field Camera 3 (WFC3) on Hubble
 - Ultraviolet imager
- Also measures less noise
 - PTC slope ratio
 - Expected ~ 1.7 @ 200 nm
 - Measured 1.09 @ 200 nm
 - SNR is $\sim 25\%$ better than expected

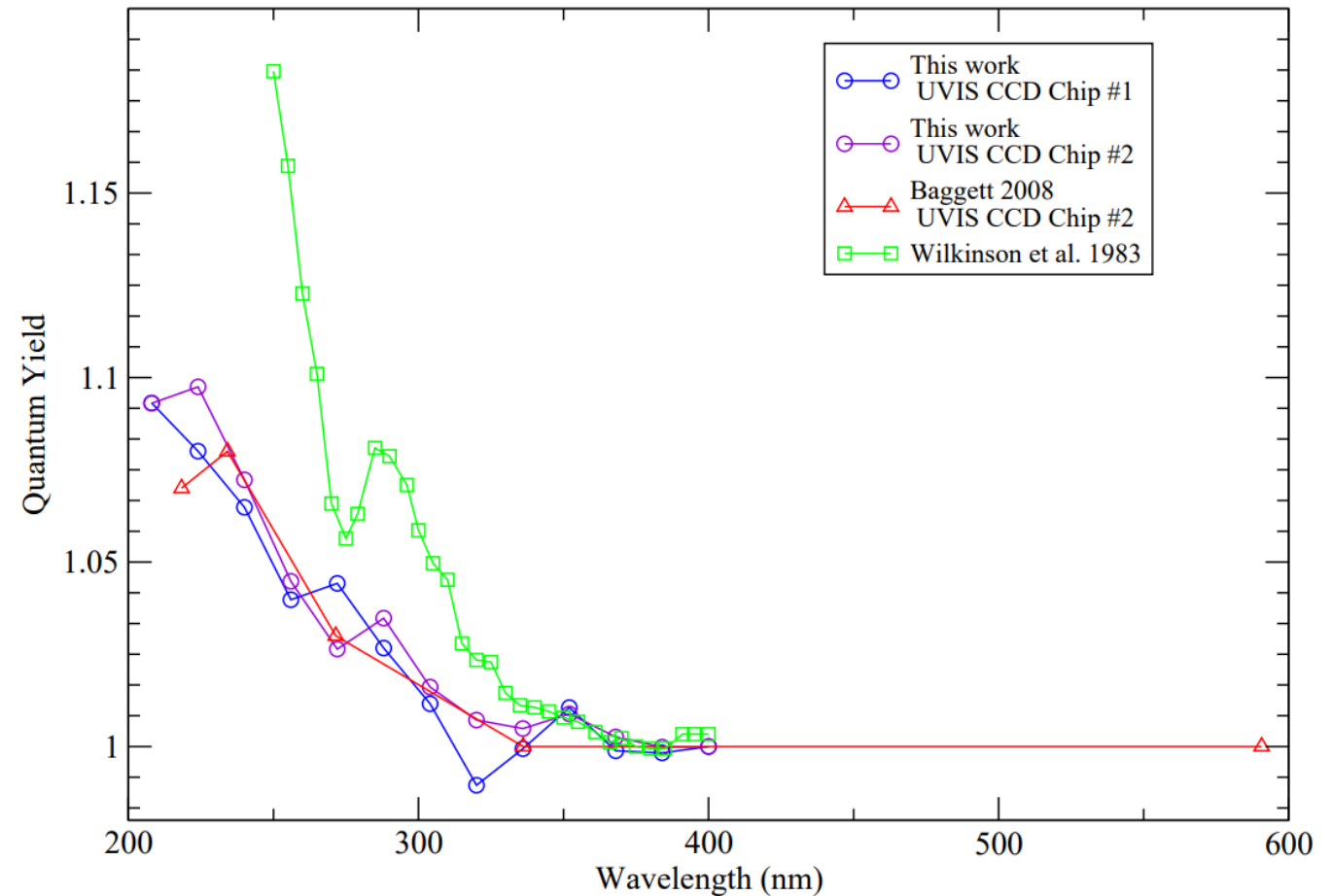
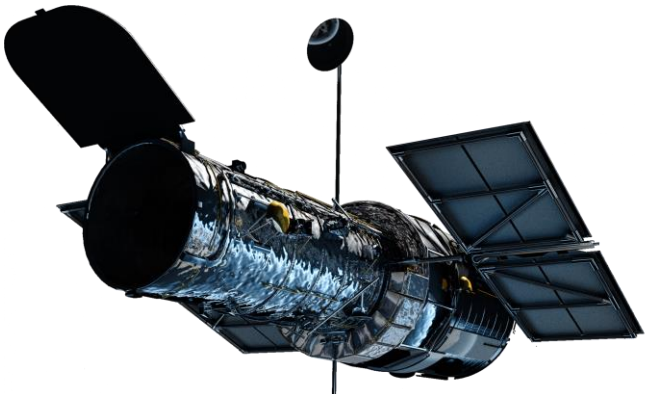
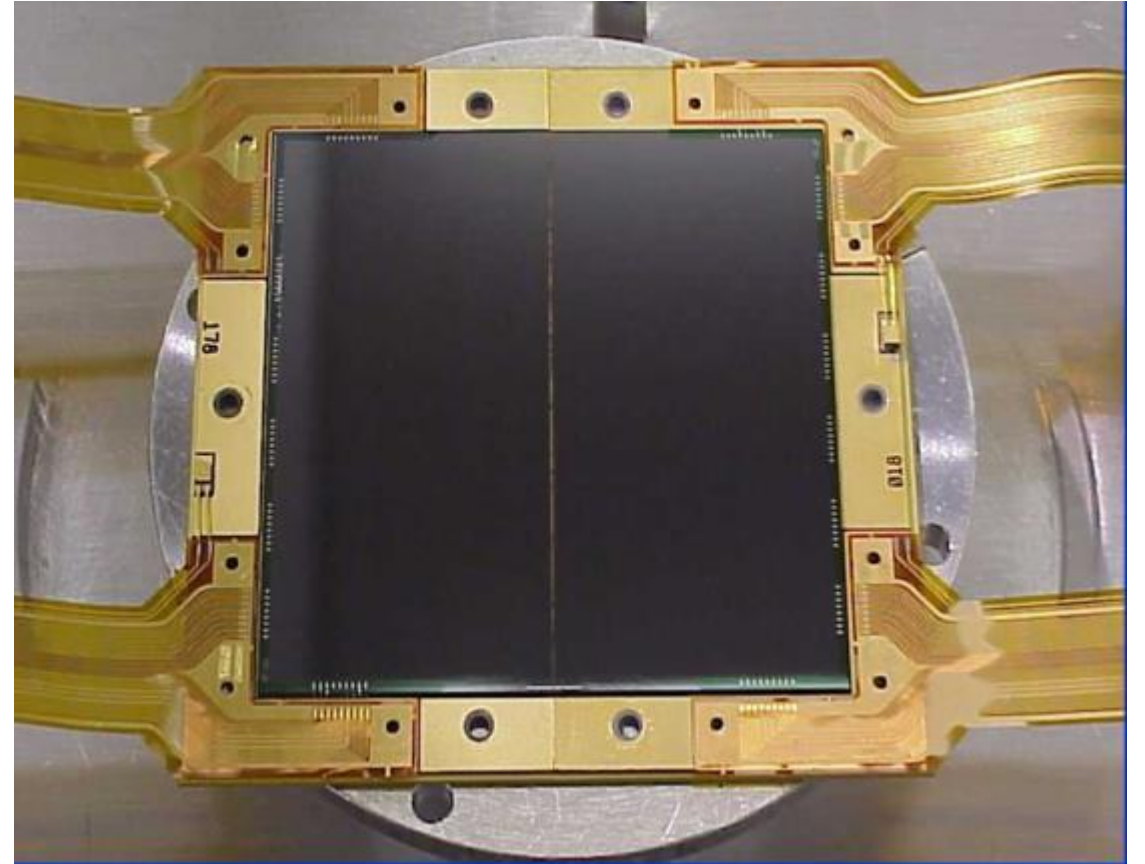


Figure 1. Our resulting quantum yields are plotted as a function of wavelength (blue and purple); the Baggett 2008 measurements are plotted in red and the quantum yield measurements of Si I by Wilkinson 1983 results are shown in green.

Outline

- Signal (Stern et al. 1994)
 - Absorbance
 - Quantum yield
 - Charge-collection efficiency
- Noise
 - Simple noise model (Stern et al. 1986)
 - Improved noise model
 - Shot noise
 - Fano noise
 - Partial charge collection noise
- Charge Spreading
- Results

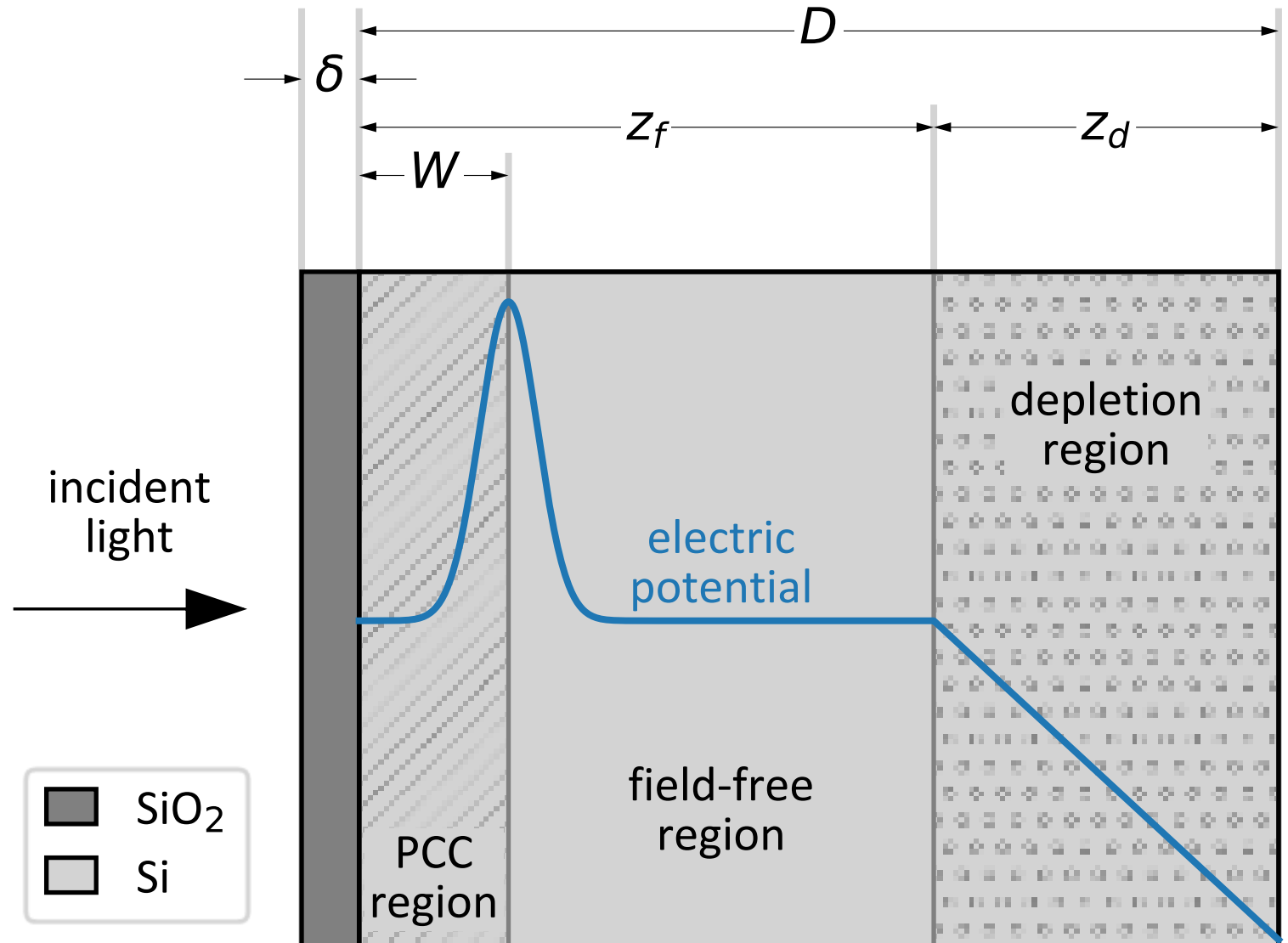


Detector package similar to the WFC3/UVIS flight detector.

Signal

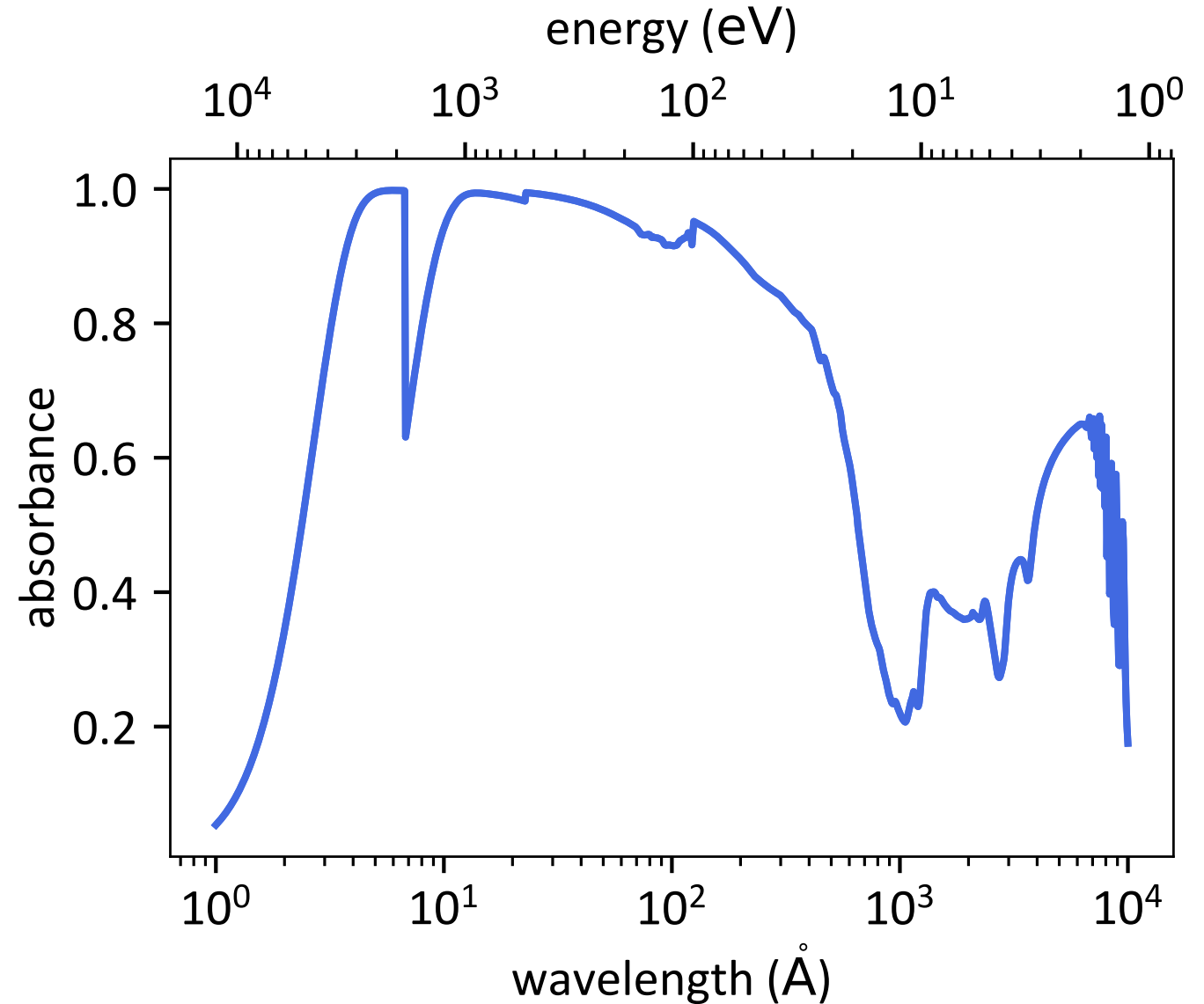
Back-illuminated Sensor Model

- Thin layer of silicon dioxide (~ 1 nm)
- Light-sensitive silicon ($16\text{ }\mu\text{m}$)
 - Partial charge collection (PCC) region (~ 100 nm)
 - Field-free region
 - Depletion region ($8\text{ }\mu\text{m}$)



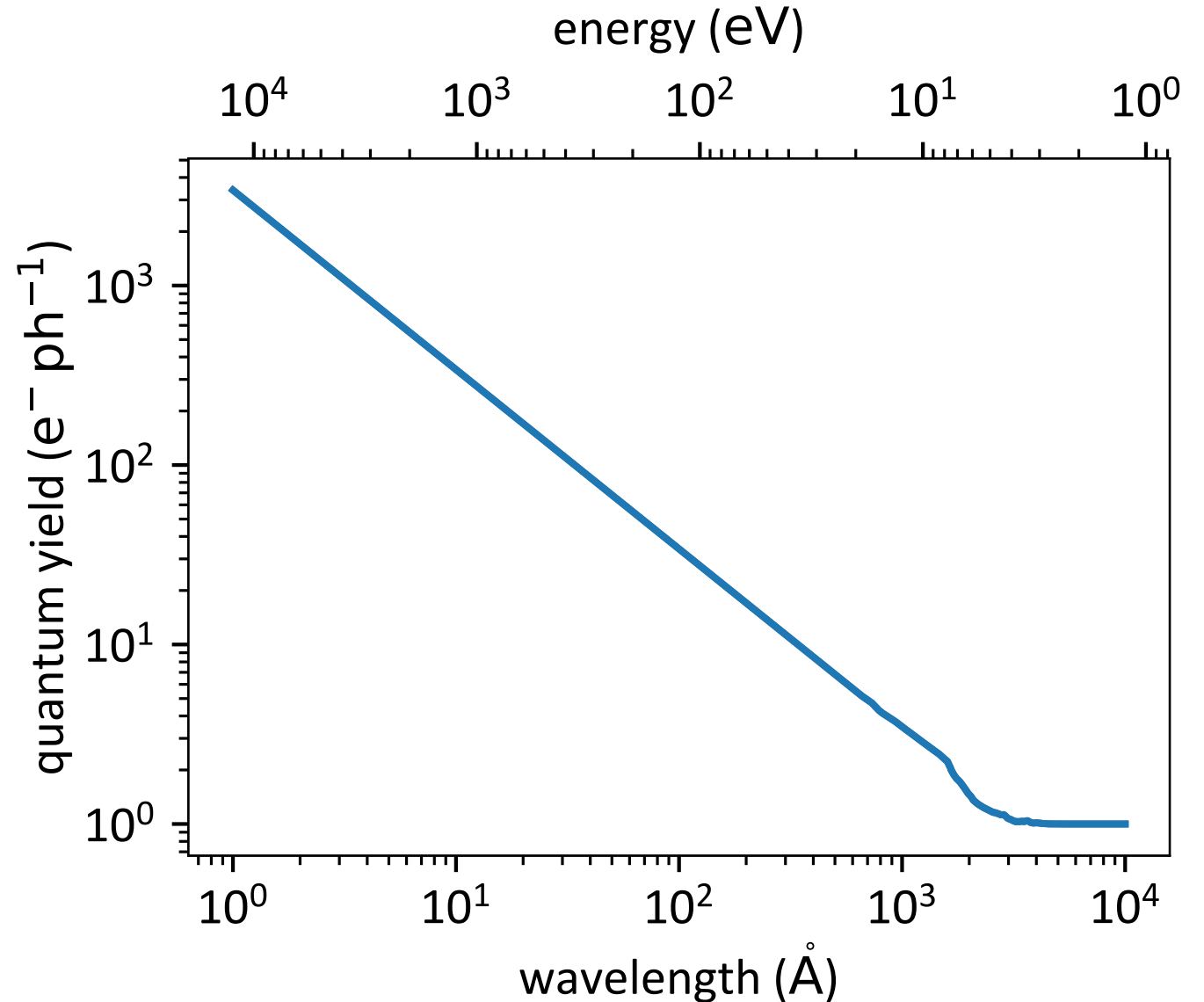
Absorbance

- Energy absorbed by light-sensitive layer
 - Reflections from each interface
 - Oxide absorption
 - Transmittance through entire sensor
- Computed using standard optical constants
 - Henke et al. (1993)
 - Palik (1997)



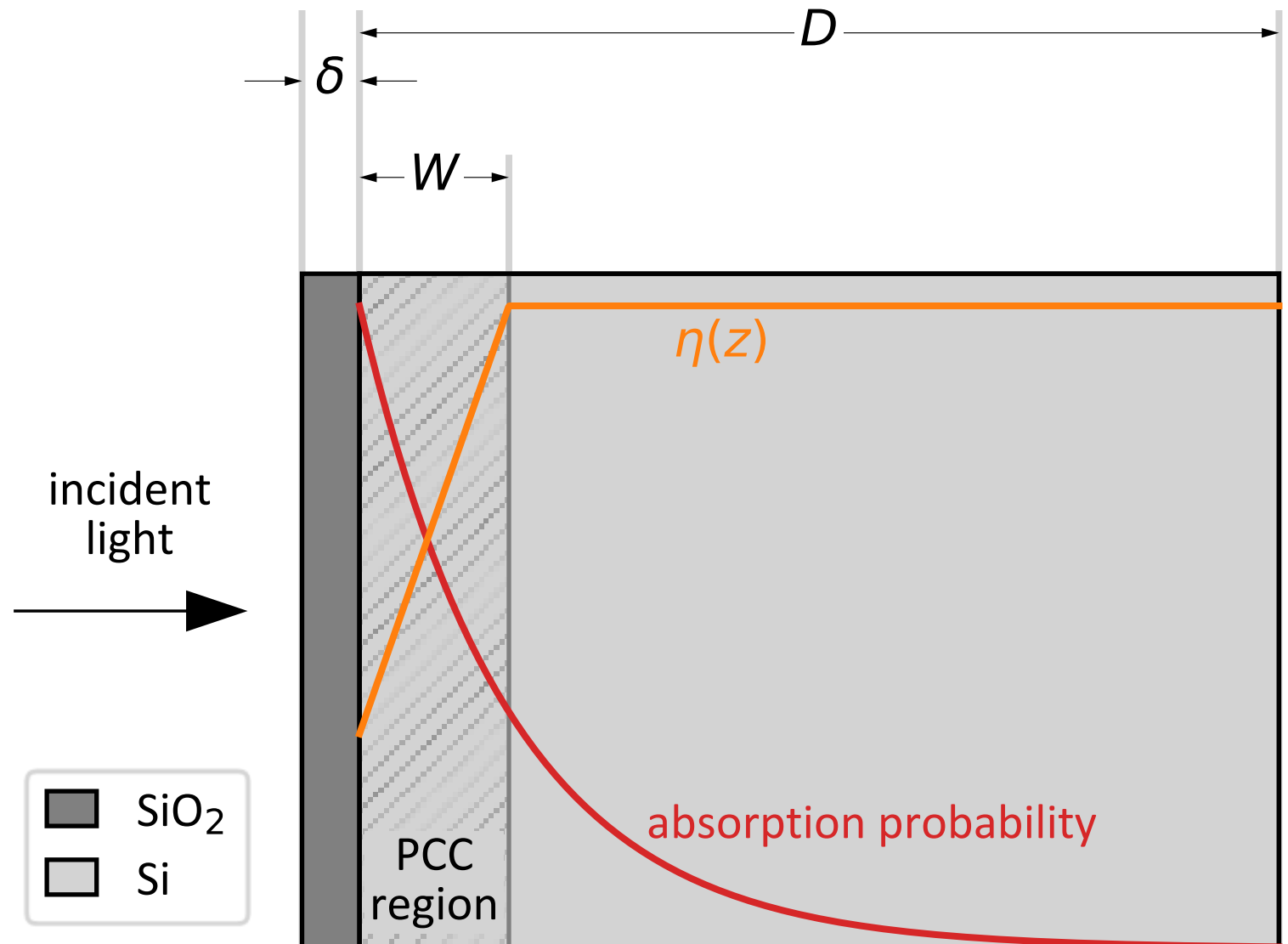
Quantum Yield

- Quantum yield (QY) is the number of electron-hole pairs generated per photon
- QY increases as photon energy increases
 - $n \approx E_\gamma / 3.65 \text{ eV}$
- Ramanathan & Kurinsky (2020) QY model



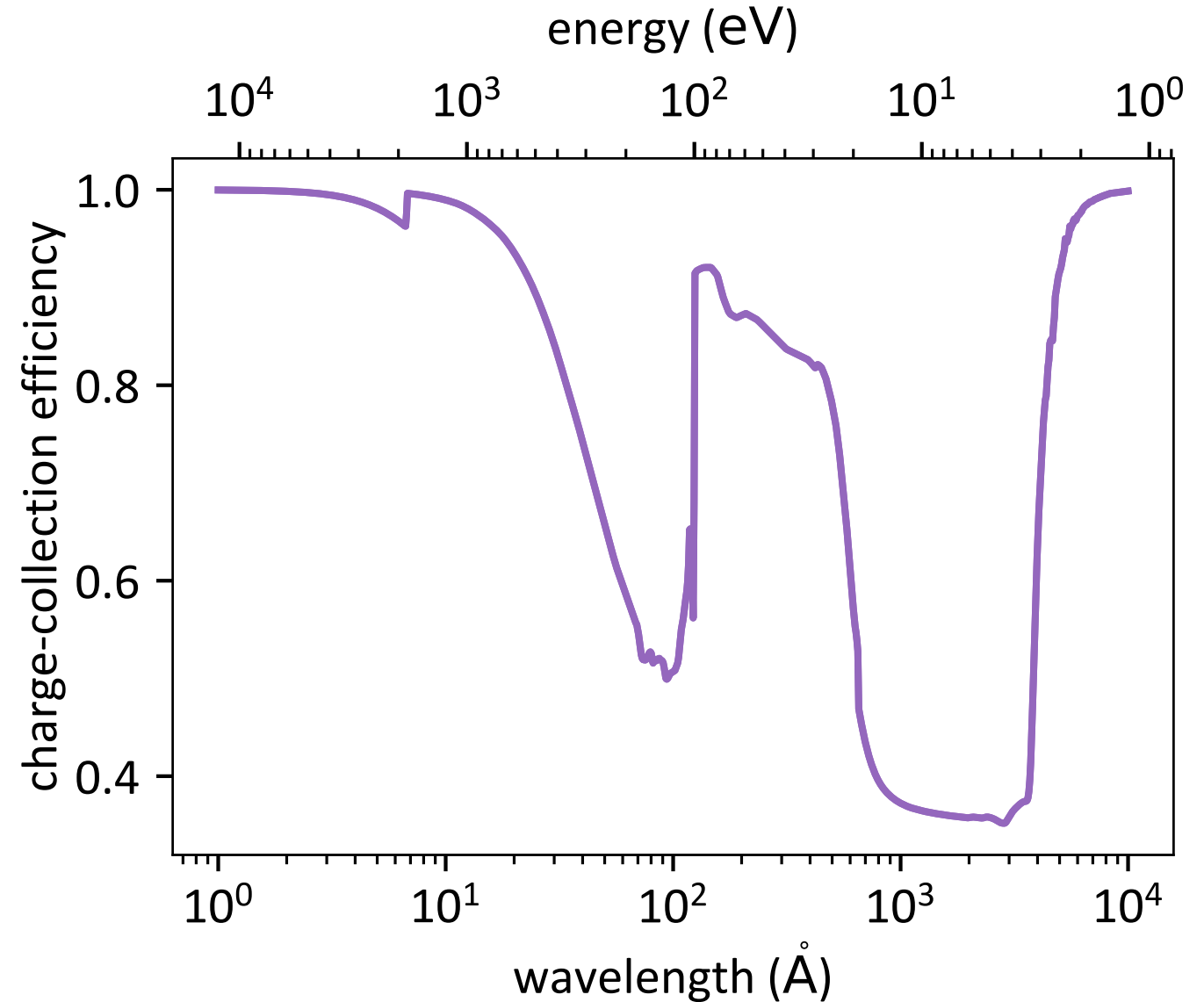
Partial Charge Collection

- Photons are absorbed according to the Beer-Lambert law (red)
- Photons absorbed in PCC region lose some electrons
- Charge-collection efficiency (CCE) is the fraction of charge measured.
- Piecewise-linear differential CCE introduced by Stern et al. (1994)
 - Denoted $\eta(z)$



Average Charge-Collection Efficiency

- Integrate Beer-Lambert law against $\eta(z)$
- Average fraction of electrons that survived to be measured



Quantum Efficiency

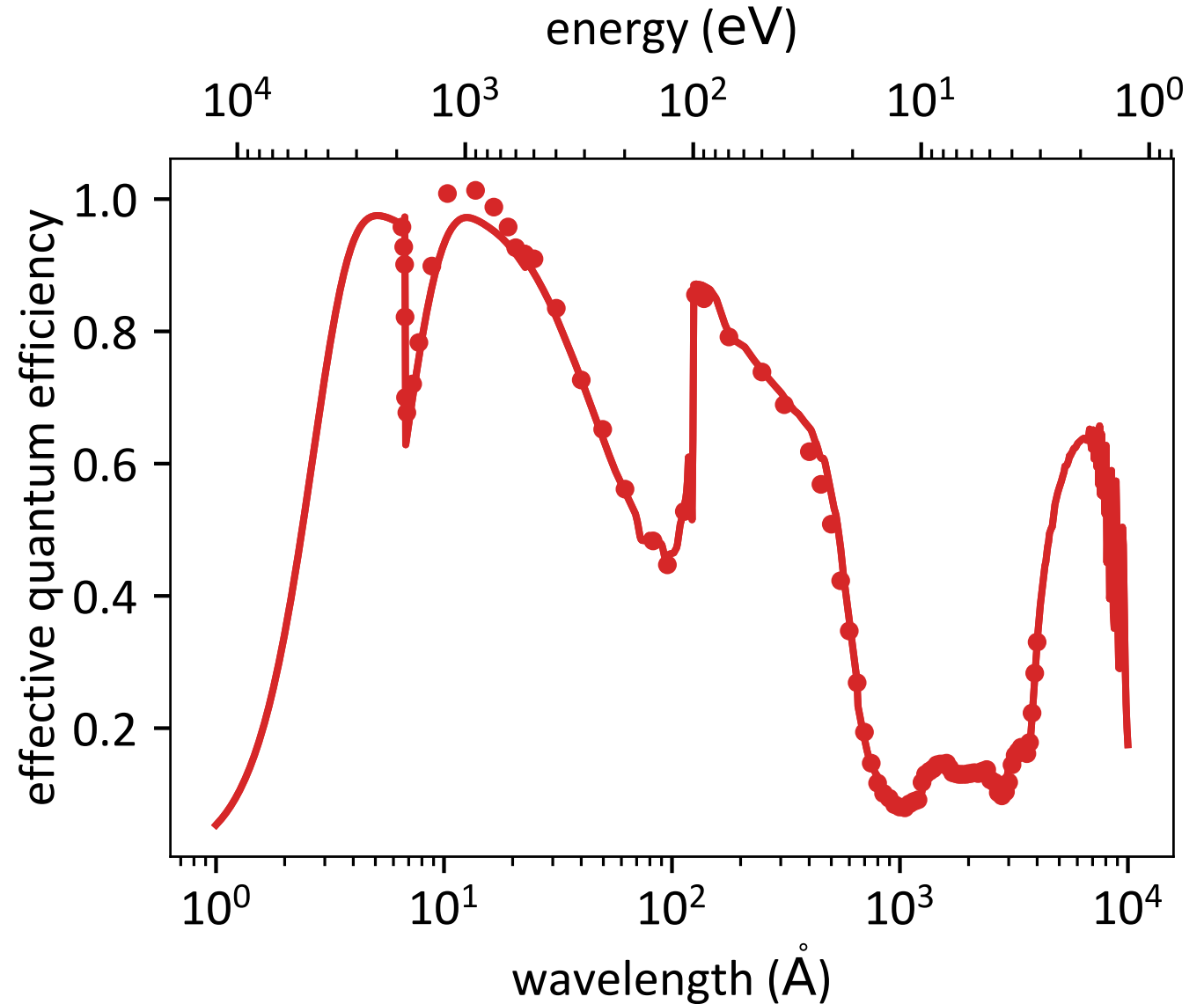
- Quantum Efficiency (QE) is the factor used to convert from incident photons to measured electrons
- Product of absorbance, quantum yield and CCE
- Allowed to be larger than one!

$$\text{QE} \equiv \left\langle \frac{N_e}{N_\gamma} \right\rangle$$
$$= A \times n \times \text{CCE}$$

Effective Quantum Efficiency (EQE)

$$\text{EQE} = \text{CCE} \times \text{Absorbance}$$

- Measured quantity
 - Compare sensor to NIST photodiode
- Between 0 and 1
- Often used to compute QE



Noise

Variance-to-mean ratio

$$\text{VMR}(X) = \frac{\text{Var}(X)}{\langle X \rangle}$$

- Convenient measure of noise
 - If X is Poisson-distributed, $\text{VMR}(X) = 1$
- Constant vs. signal, unlike SNR
- Equal to the slope of the photon-transfer curve

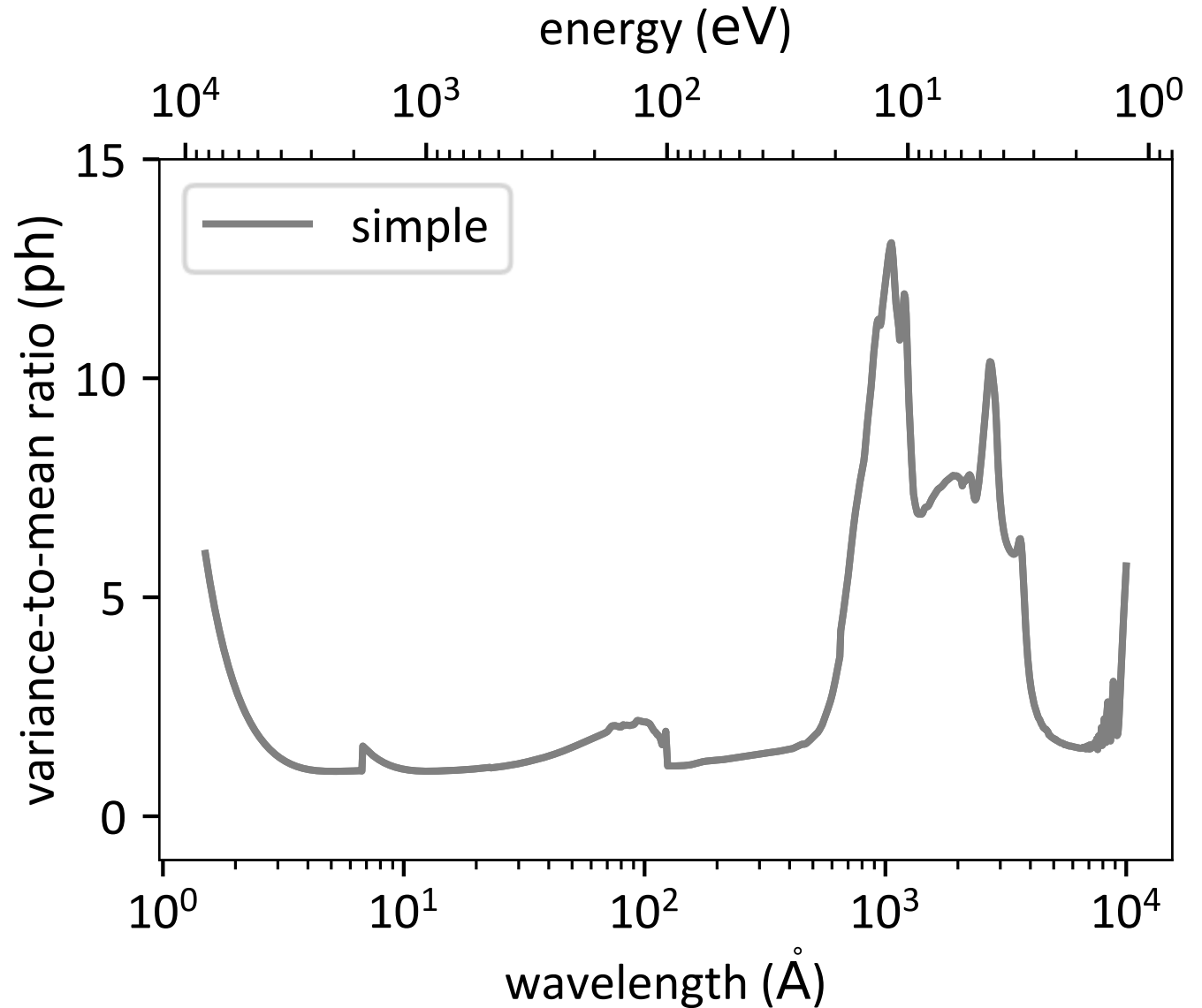
Simple Noise Model

- Used by IRIS and WFC3
- Number of measured photons proportional to EQE
- All-or-nothing charge collection

$$\langle N_{\gamma}'' \rangle = \text{EQE} \times \langle N_{\gamma} \rangle$$

$$\sigma_{N_{\gamma}} = \sqrt{\text{EQE} \times \langle N_{\gamma} \rangle}$$

$$\Rightarrow \text{VMR}(N_{\gamma}) = \frac{1}{\text{EQE}}$$



An Improved Noise Model

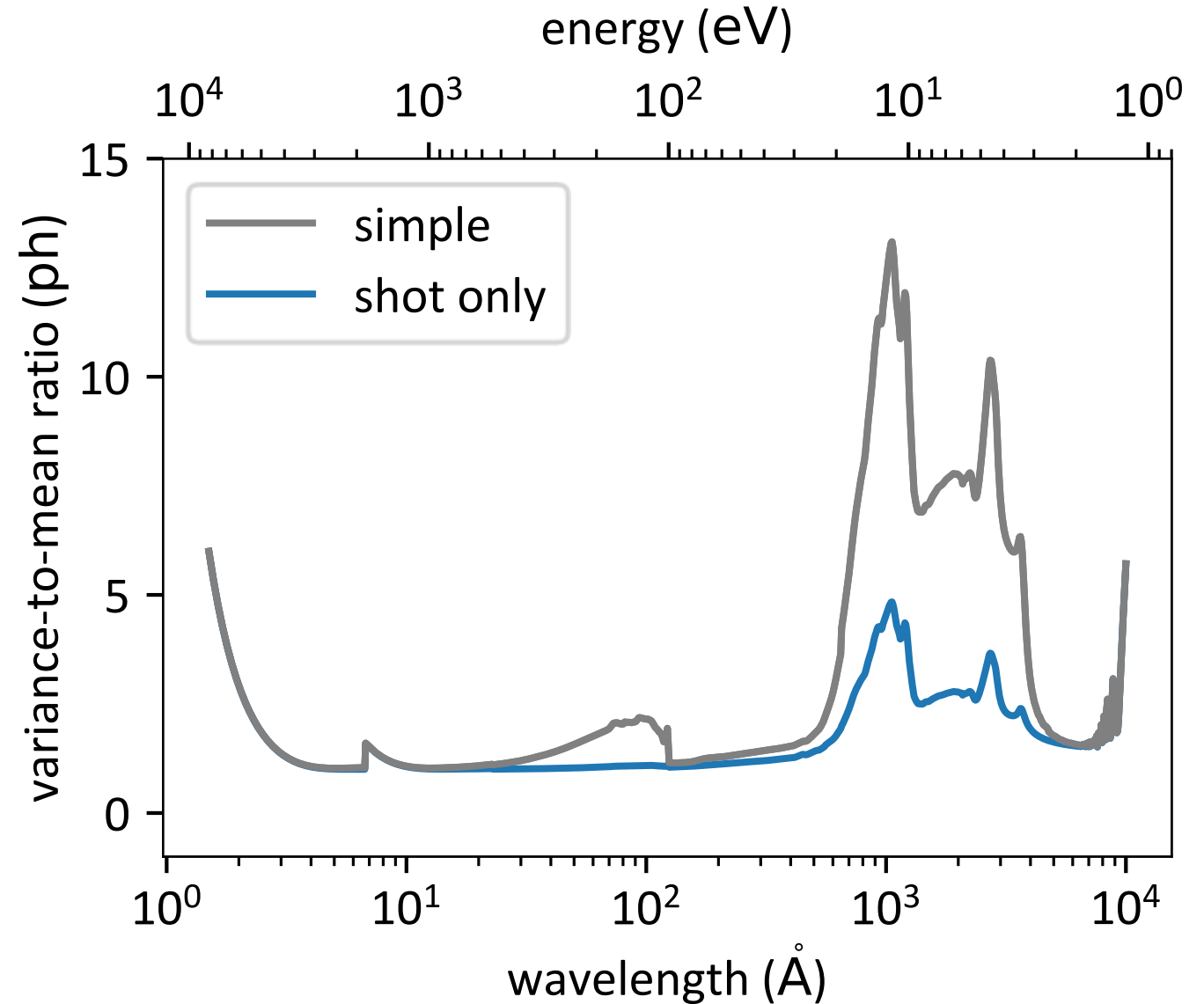
- Shot noise
- Fano noise
 - Inherent randomness in the charge-generation process
- **PCC noise**
 - Uncertain CCE due to random absorption depth

Shot Noise

- Absorbed photons are Poisson-distributed

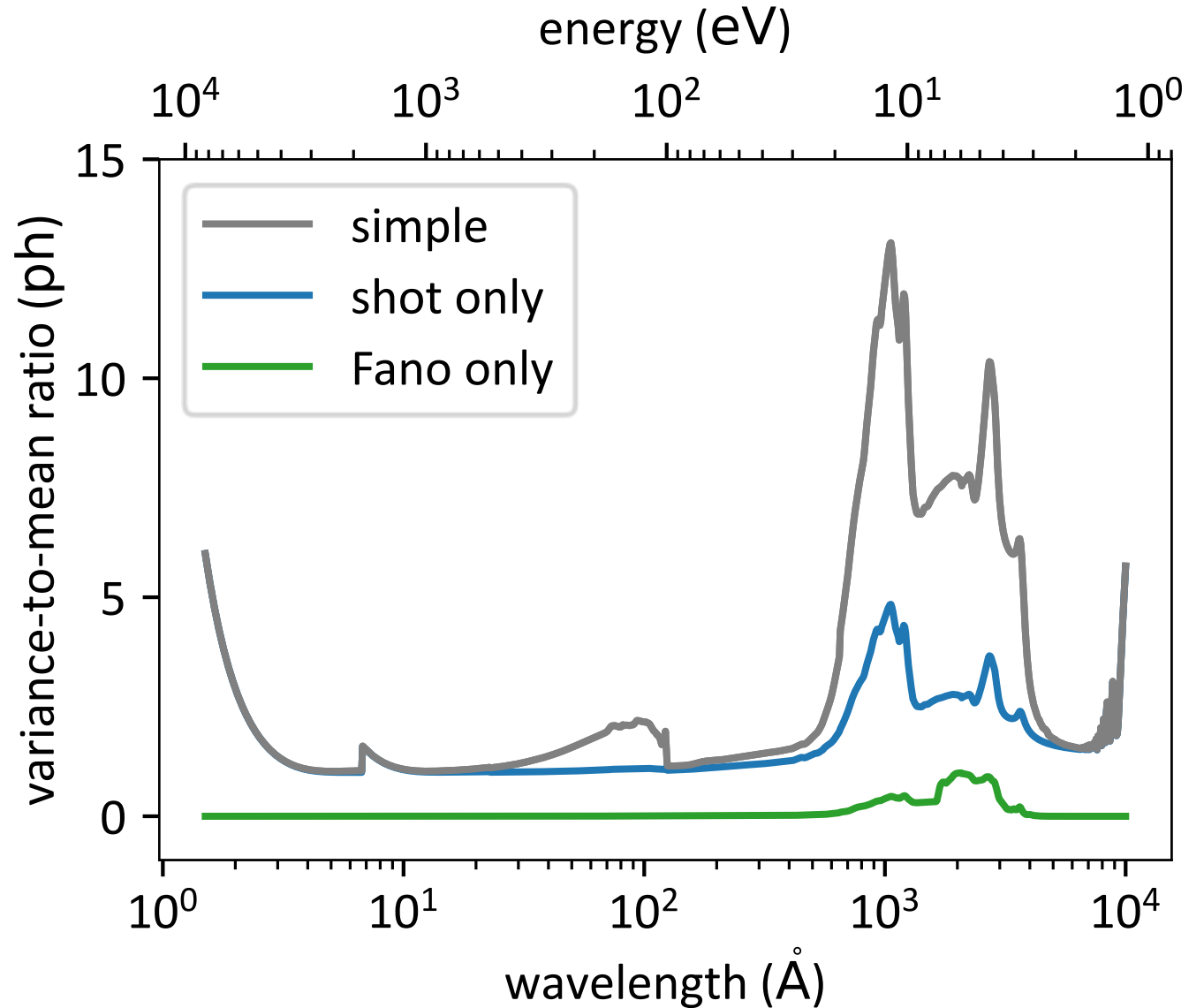
$$N'_\gamma \sim \text{Poisson}(A \times \langle N_\gamma \rangle)$$

$$\text{VMR}(N_\gamma) = \frac{1}{A}$$



Fano Noise

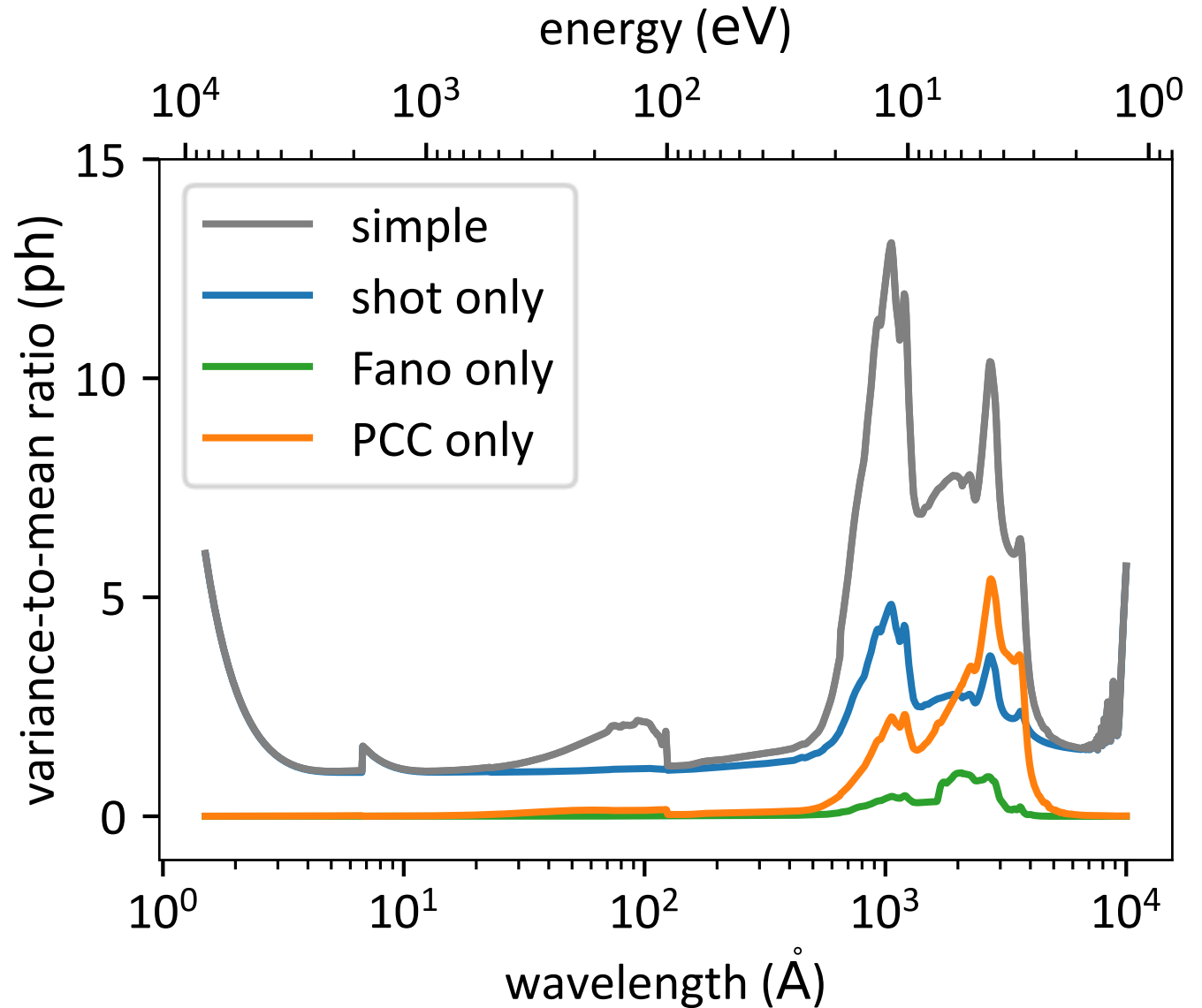
- Quantum yield uncertainty
- Described by Fano factor
 - $\mathcal{F} = \text{VMR}(\text{QY}) \simeq 0.1$
- Ramanathan & Kurinksy (2020) QY model
- Small compared to other noise sources



PCC Noise

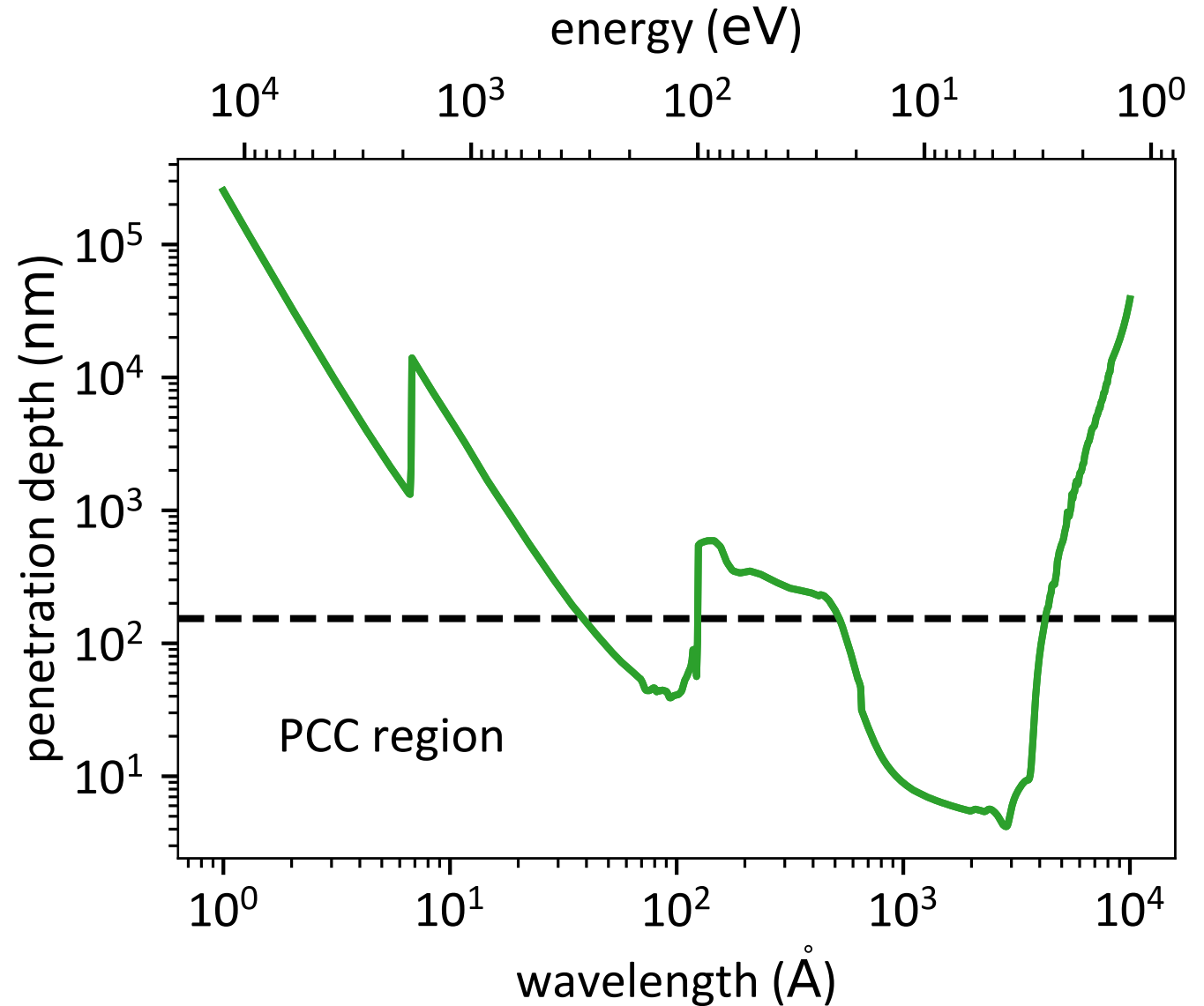
- Since each photon is absorbed at a different depth, each experiences a different CCE.
- Each electron has a chance of being absorbed

$$N_e = \sum_{i=0}^{N'_\gamma} \text{Binomial}(n_i, \eta(z_i))$$



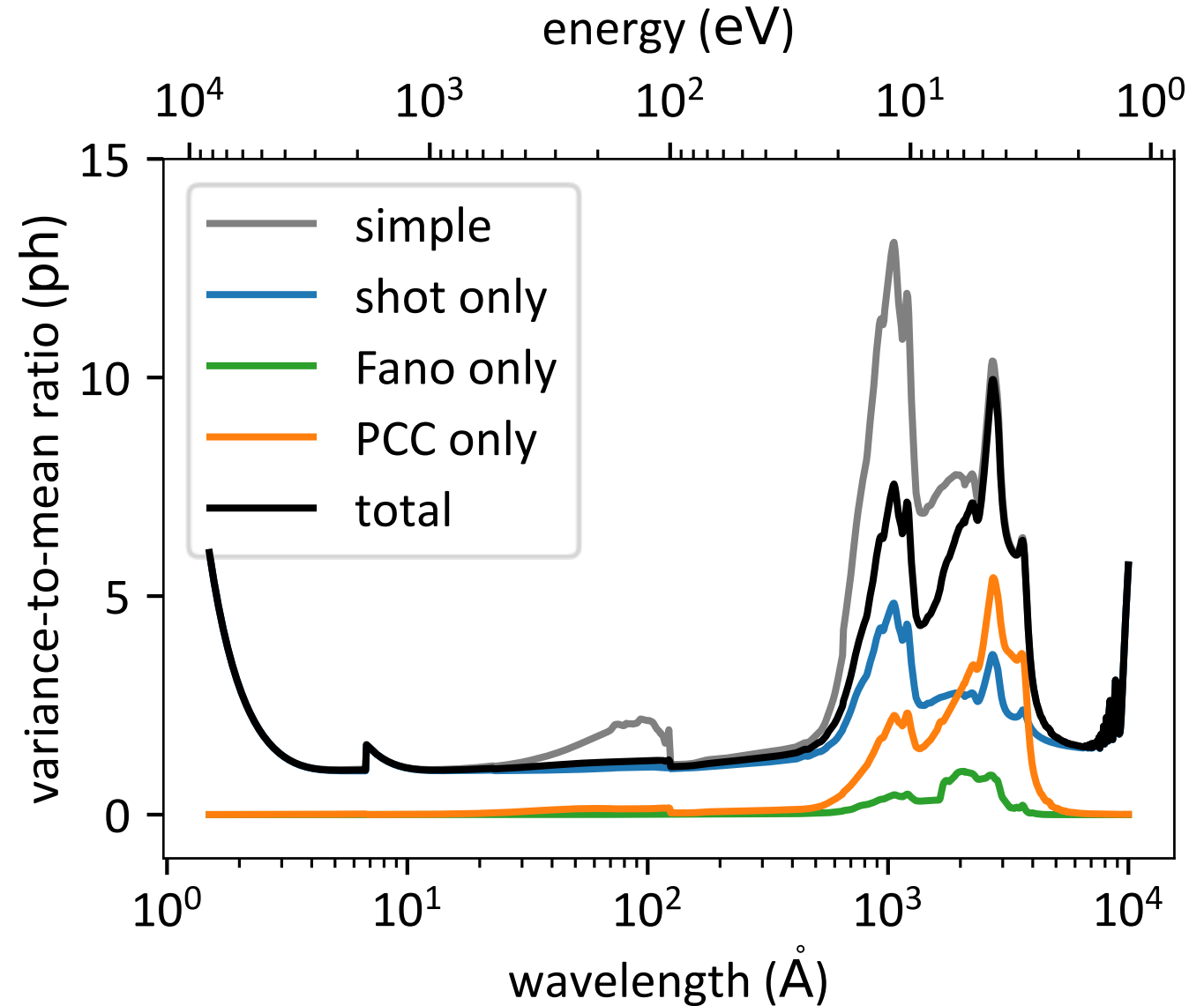
Penetration depth

- Some ultraviolet wavelengths have a penetration depth smaller than the PCC region
 - $\sim 40 - 100 \text{ \AA}$
 - $\sim 500 - 4500 \text{ \AA}$
- PCC noise is important in these regions



Total Noise

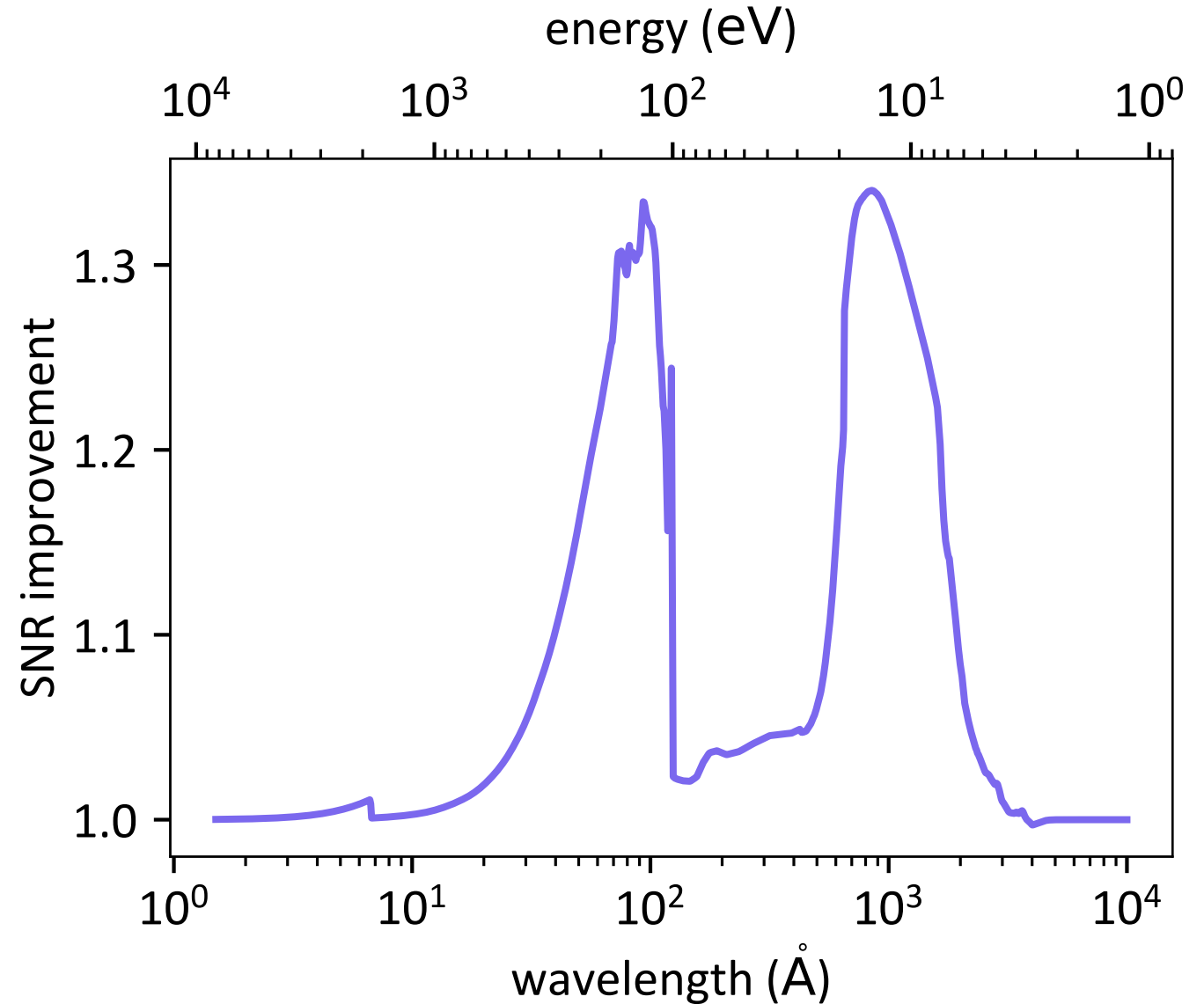
- Sum of shot, Fano, and PCC noise
- VMR of improved model is smaller than the simple model in the UV
 - Shallow penetration depth
- The partial events improve the noise performance
 - More photons are detected



SNR improvement

$$= \sqrt{\frac{\text{simple VMR}}{\text{total VMR}}}$$

- SNR improves by up to ~30% compared to the simple model



Charge spreading?

- Charge spreading is when electrons diffuse into adjacent pixels.
- Proportional to size of field-free region
- Could explain the discrepancy since this increases the correlations between pixels

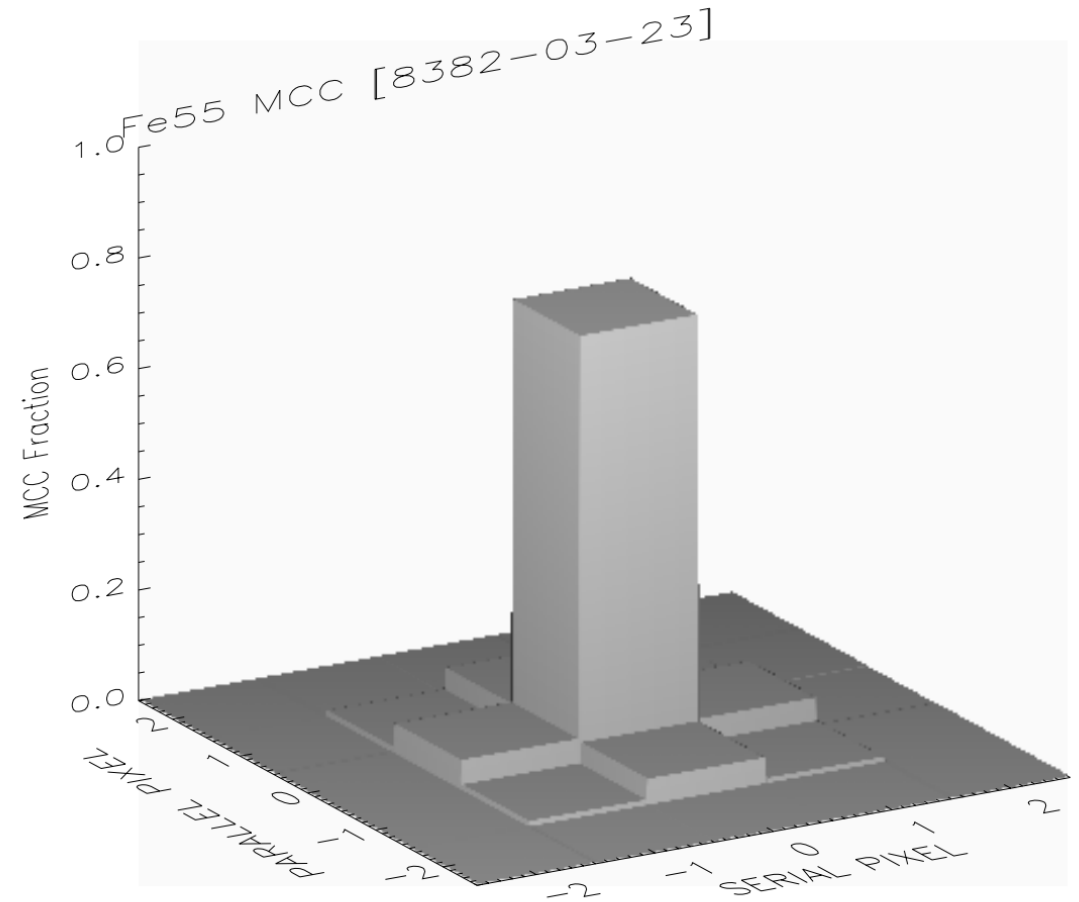
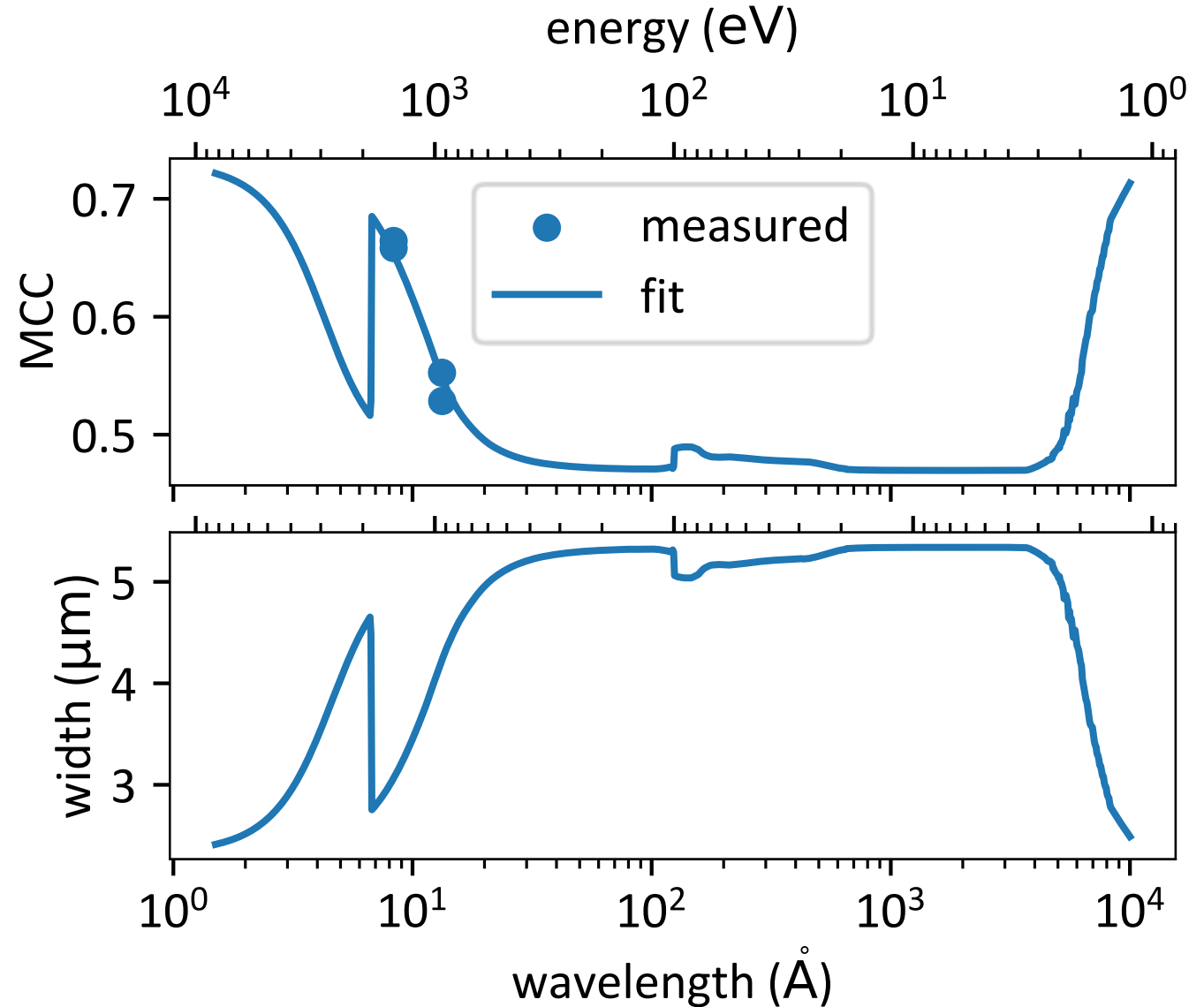


Figure 13. ^{55}Fe (5.9 keV) single photon PSF. MCC is 0.78 for central pixel.

Stern et al. (2003)

Charge Spreading Model

- Janesick (2001) charge spreading model
- Stern et al. (2004) measurements
- Fit size of field-free region
- Predicts a similar amount of charge spreading for both UV and visible light
 - Cannot explain IRIS and WFC3 discrepancies



Compare to Measurements

Instrument	λ_{UV} (Å)	λ_{vis} (Å)	model	measurement
IRIS	1350	4500	1.57	1.50
WFC3	2080	4000	1.14	1.09
	2240	4000	1.11	1.09
	2400	4000	1.08	1.07
	2560	4000	1.06	1.04
	2720	4000	1.04	1.04
	2880	4000	1.03	1.03
	3040	4000	1.01	1.01

Affected Instruments

- AIA
 - 94 Å
 - 1600 Å
- IRIS
 - 1330 Å
 - 1400 Å
- MUSE
 - 108 Å

Instrument	λ (Å)	VMR (ph)	VMR (e ⁻)	improvement
AIA	94	1.23	20.14	1.34
	131	1.10	24.69	1.02
	171	1.16	18.44	1.03
	193	1.18	16.22	1.04
	211	1.20	14.88	1.04
	304	1.30	10.15	1.05
	335	1.33	9.20	1.05
	1600	4.92	1.47	1.23
	1700	5.55	1.38	1.18
IRIS	1330	4.41	1.65	1.28
	1400	4.36	1.60	1.27
	2796	9.68	1.07	1.03
	2832	9.37	1.07	1.03
MUSE	108	1.24	19.39	1.27
	171	1.16	18.42	1.04
	284	1.28	10.93	1.05

Conclusions

- Partial charge collection is responsible for discrepancy in IRIS and WFC3 noise statistics
- Charge spreading is less important
- The model predicts better SNR in some wavelengths than expected from the simple model
- Results affect AIA, IRIS, WFC3, and MUSE.
- Code provided in our `optika` Python package.
 - Published on PyPI